

Titanic Dataset Analysis Report

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1. About the Dataset

The Titanic dataset contains details about passengers on the Titanic ship. It includes information such as name, age, sex, ticket class, fare, port of embarkation, and whether they survived. This dataset is helpful for learning how to work with real-world data, especially when dealing with missing values and categorical information.

2. Dataset Analysis

a. Loading the Data

I used the panda's library to load the dataset from the file train.csv. After loading it, I checked the first few rows, the column types, and some statistics using `.head()`, `.info()`, and `.describe()`

b. Handling Missing Data

I found some missing values in the columns:

- Age: Filled with the median age.
- Embarked: Filled with the most common value (mode).
- Cabin: Had many missing entries, so I replaced them with the string "Unknown".

These steps helped make the dataset more complete and ready for analysis.

c. Data Visualization

To understand patterns in the data, I created the following charts using matplotlib:

- A bar chart showing survival by gender (more women survived than men).
- A chart showing survival by ticket class (first-class passengers had higher survival rates).
- A histogram showing the age distribution of passengers.
- A bar chart showing survival by embarkation port.

d. Exploring the Data

I used the `groupby()` function to explore the data more:

- ✓ I calculated the average age for males and females.
- ✓ I looked at survival rates by both gender and ticket class.
- ✓ I also calculated statistics like average fare by gender and class.

To better understand survival rates, I used `unstack()` to create a simple table, and then showed this data as a heatmap using `imshow()`.

Conclusion

This project helped me practice key data analysis skills like handling missing data, using `groupby()`, creating visualizations, and understanding patterns in real data. I also learned how factors like gender and ticket class influenced survival on the Titanic.